



ABHAY PRATAP SINGH

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ABOUT ME

I am passionate about building user-centric web applications and have a strong interest in combining data science with real-world applications, particularly in areas such as machine learning, computer vision, and intelligent systems. I enjoy working across both frontend and backend technologies to develop scalable and impactful solutions.

WORK EXPERIENCE

CNH Industrial – Gurugram, India

Address: 12th Floor, Tower B, Building 14, DLF Cyber City, DLF Phase 2, Sector 24, Gurugram, Haryana , 122010 Gurugram (India)

Frontend Developer

[04/04/2024 – 04/03/2025]

- Developed a scalable platform for configuring CAN, Duty, MH, and Universal profiles, improving configuration efficiency by ~30% and reducing manual setup time across multiple systems.
- Implemented CAN parameter management for AM53, PCM, and CM1X systems, reducing configuration errors by ~25% and improving overall system reliability in production environments.
- Designed and optimised PostgreSQL database operations, improving query performance by ~20% and accelerating data retrieval for high-volume configuration data.
- Built responsive and user-friendly web applications using HTML, CSS, JavaScript, Angular, and Bootstrap, reducing page load time by 15–20% and enhancing cross-device usability.
- Integrated RESTful APIs using JSON/XML and validated endpoints with Postman, reducing integration issues by ~20% and ensuring seamless backend communication.
- Collaborated with cross-functional teams using GitHub for version control, improving deployment efficiency and minimizing code conflicts during development cycles.
- Developed and executed unit test cases using Karma.js, increasing code coverage and reducing production bugs by ~30%.
- **Contributed to product release cycles**, participating in feature validation, bug fixing, and deployment support, ensuring timely and high-quality product delivery.

Two waits Technologies – Gurugram, India

Web developer (Internship)

[15/03/2022 – 14/04/2022]

- Designed and developed responsive web applications using HTML, CSS, and JavaScript, improving user engagement by ~25%.
- Optimized website performance, reducing page load time by ~20% through efficient coding and asset optimization.
- Built intuitive and user-friendly UI components, increasing user retention and improving overall user experience.
- Conducted cross-browser testing (Chrome, Edge), ensuring 100% compatibility and reducing UI bugs by ~40%.
- Identified and fixed front-end issues through debugging, improving application stability and performance.
- Collaborated with team members to deliver features on time, improving development efficiency.
- Provided technical support to 50+ users, resolving ~90% of issues within the first response.
- Assisted in troubleshooting website-related problems, enhancing customer satisfaction and usability.
- Implemented responsive design techniques, ensuring seamless experience across mobile, tablet, and desktop devices.

EDUCATION AND TRAINING

Master of Science (M.Sc)

University of Parma [15/09/2025 – Current]

City: Parma | **Country:** Italy | **Website:** <https://www.unipr.it/en> | **Field(s) of study:** Data Science For Management

Bachelor of Technology (B.Tech.)

Parul University [25/05/2019 – 25/03/2023]

City: Vadodara | Country: India | Website: <https://paruluniversity.ac.in/> | Field(s) of study: Computer Science and Engineering

LANGUAGE SKILLS

Mother tongue(s): Hindi

Other language(s):

English

LISTENING B2 READING B2 WRITING C1

SPOKEN PRODUCTION B2 SPOKEN INTERACTION B2

Italian

LISTENING A1 READING A1 WRITING A1

SPOKEN PRODUCTION A1 SPOKEN INTERACTION A1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Technical Skills

HTML5 / CSS / Bootstrap / JavaScript / Python / Object-Oriented Programming / MySQL / Machine learning / Angular15 / KarmaJS / Figma / XML/JSON / Web API / Postman / Github / Git

PROJECTS

[20/02/2026 – Current]

MusicEra – Emotion-Based Music Recommendation System

- Designed and developed an intelligent web application that **captures user emotions through real-time facial expression analysis** and delivers **personalised music recommendations** based on detected mood.
- Engineered a responsive and intuitive frontend using **HTML, CSS, and JavaScript**, ensuring seamless user interaction and cross-device compatibility.
- Built a robust backend in **Python**, leveraging **OpenCV** for real-time webcam video capture, image preprocessing, and facial feature extraction.
- Implemented **deep learning-based emotion recognition** using **DeepFace** and **TensorFlow**, accurately classifying emotional states such as happiness, sadness, anger, and neutrality with **high precision (~90%)**.
- Developed a real-time data processing pipeline to map detected emotions to **curated music playlists dynamically**, enhancing personalised music streaming experiences.
- Optimised **frontend-backend communication**, enabling **low-latency, real-time emotion-to-music recommendations**.
- Applied best practices in modular coding, API integration, and environment configuration to ensure **scalability, maintainability, and extensibility**.
- Enhanced user engagement and satisfaction by providing **context-aware and mood-based music recommendations**, promoting a unique and immersive listening experience.

Plant Disease Detection

- Designed and developed a comprehensive web application to identify plant diseases by analysing uploaded leaf images, leveraging **machine learning algorithms and advanced image recognition techniques**.
- Built a **user-friendly interface** allowing gardeners, farmers, and plant enthusiasts to easily upload photos and receive **accurate disease diagnoses with suggested treatments**, improving plant care efficiency.
- Integrated a **dataset of 16,000+ distinct plant leaf images** (sourced from Kaggle), enabling robust training of models and increasing diagnostic accuracy.
- Implemented **real-time image analysis**, delivering disease identification results in seconds, streamlining the diagnostic process for end users.
- Optimised machine learning workflows and backend operations, ensuring **fast response times and scalable performance** for multiple simultaneous users.
- Documented findings and insights for each detected disease, enhancing user understanding and enabling effective plant health management.
- Promoted hands-on experimentation and learning in plant pathology, providing a practical tool for **researchers, students, and hobbyists**.